Location, location, location: does a tabular foundation model need spatial features?

TabPFN-3.5 against spatial econometrics and gradient boosting on 130,000 declared transaction prices in São Paulo

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Abstract

Mass appraisal models encode the dependence of property prices on location explicitly, through spatial weights matrices, spatial lags, geographically weighted local models or engineered neighbourhood features. We ask whether a tabular foundation model can do without them. TabPFN-3.5 receives the property attributes and the geographic coordinates as two ordinary numeric columns, with no spatial modelling of any kind, and is compared with an OLS hedonic model, a spatial autoregressive lag model (SAR, GM estimation), XGBoost with a k-nearest-neighbour spatial lag, rotated coordinates and nested tuning, and a geographically weighted XGBoost. The data are 82,187 declared transaction prices from the São Paulo real-estate transfer tax (ITBI) in 2025 and 47,810 in the first seven months of 2026, for apartments, houses and commercial units. The evaluation has two legs: leave-one-block-out cross-validation over ten spatial blocks, and an out-of-time test that fits on 2025 and predicts 2026. Refutation criteria were fixed in advance: worse pooled error, losses in most blocks under paired tests, or more residual spatial autocorrelation (Moran's I) than the explicitly spatial baselines. Out of time, the plain-table model beats all three spatial specialists in at least 9 of 10 blocks (RMSE 0.265 in log unit price against 0.283–0.326; MAPE 19.5% against 20.9–25.0%) and leaves the least residual autocorrelation of the models compared ($I = 0.092$); with the raw postal code in place of the coordinates — one text column, no geometry — every one of those verdicts holds. Across unseen blocks it ties both XGBoost variants on error and beats SAR, but leaves more spatial structure in its residuals ($I = 0.348$ against 0.309–0.326), a partial refutation, on the record. Permutation SHAP shows the two coordinates as the model's strongest inputs; the postal-code result shows that what they carry, inside an observed market, is location as an identifier rather than geometry, and that only coordinates extrapolate to districts without sales. For mass appraisal inside a market with a sales history, a foundation model given the raw table is the strongest predictor in this comparison, with one fit of fifteen seconds and no spatial engineering.

Keywords: mass appraisal; tabular foundation models; TabPFN; spatial autoregression; spatial cross-validation; hedonic prices; São Paulo.

1 Introduction

Mass appraisal at city scale rests on the hedonic view of the property market [21]: the price of a heterogeneous good is a function of its attributes, and location is the attribute that dominates the others — near things are more related than distant things [23]. Prices of neighbouring properties are therefore not independent observations, which breaks the assumption behind the

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ordinary least squares regression that appraisal practice relies on [2]. The established answers model the dependence explicitly: a spatial weights matrix and an autoregressive term in spatial econometrics [2, 12], local models re-fitted across the map in geographically weighted regression [7] and its tree-based descendants [8], or neighbourhood features hand-fed to a gradient-boosting learner. All of it works, and all of it is engineering that has to be redone for every city and every dataset: a weights matrix to build, a bandwidth to tune, a lag to recompute inside every training fold.

Tabular foundation models offer a different bargain. TabPFN is a transformer [24] pre-trained on millions of synthetic tabular tasks; at use time it performs in-context learning, taking the whole training table as its prompt and predicting the test rows in one forward pass, with no gradient steps and no tuning [9, 10]. The third generation extended the approach to large tables and regression [17], and version 3.5 reads high-cardinality categorical and free-text columns natively [18]. Whether such a model, handed latitude and longitude as two plain numeric columns, learns the spatial dependence that the specialists encode by construction is an open empirical question — and a practical one, because if it does, the spatial machinery above becomes optional for a large class of valuation problems.

The question has a concrete origin. In a comparative study on 5,005 apartment listings in Belo Horizonte, Brazil, TabPFN v3 with raw coordinates outperformed OLS, SAR, SEM and a semiparametric spatial lag model under spatial block cross-validation [3]. That result invited a harder test, and this paper is that test: declared transaction prices instead of listings, three property types instead of one, 82,187 training rows instead of 5,005, a one-year-ahead leg on top of the spatial block cross-validation, a residual autocorrelation criterion that can refute the claim even where the error cannot, and a stronger set of spatial opponents. The framing is falsificationist: the hypothesis is stated before the runs, with the criteria that would refute it, and the comparison is deliberately asymmetric *against* it — the baselines keep every spatial advantage, the foundation model gets the plain table.

The hypothesis is stated so that it can fail:

> A tabular foundation model that receives only the property attributes and the coordinates as two plain numeric columns — no spatial weights matrix, no spatial lag, no engineered neighbourhood features — predicts spatially correlated prices as well as, or better than, specialised models that encode spatial dependence explicitly.

It is refuted if any of the following holds under a shared protocol: worse pooled RMSE or MAPE; losses in most spatial blocks under paired tests; or more spatial autocorrelation left in its out-of-fold residuals than the explicitly spatial models leave in theirs.

The paper contributes: (i) a comparison of a tabular foundation model with spatial econometric and spatially augmented tree baselines — a geographically weighted XGBoost among them — on declared prices at city scale; (ii) a two-leg protocol, spatial block cross-validation plus out-of-time testing, with pre-registered refutation criteria that include residual spatial autocorrelation; (iii) ablations that isolate what carries location inside the model, replacing the coordinates with a raw postal code and adding the free text of the tax form; (iv) a public, reproducible pipeline over open municipal data; and (v) the finding itself, the partial refutation included.

2 Related work

Spatial hedonic models. The hedonic framework [21] combined with Tobler's first law [23] leads to the spatial econometric family: the spatial lag (SAR) and spatial error models, estimated by maximum likelihood or by the generalised-moments procedure of Kelejian and Prucha [11], with the weights matrix as the carrier of the neighbourhood structure [2, 12]. Geographically weighted regression drops the global coefficient vector and re-fits a local model at every calibration location with distance-decaying weights [7].

Machine learning with spatial features. Tree ensembles do not see space unless it is fed to them. The standard devices are a spatial lag of the target computed from the training neighbours, coordinate transformations such as oblique or rotated axes [15], and distance covariates. Geographical-XGBoost [8] transplants the geographically weighted idea to gradient boosting: one local XGBoost per calibration location, blended with a global model. Evaluating any of this honestly requires spatial cross-validation, since random splits let neighbours of a test point sit in its training set [14, 20].

Tabular foundation models, and space inside them. TabPFN [9, 10] and its third generation [17, 18] replace per-dataset training with in-context learning over synthetic priors. Two recent lines bring space *to* the model: TabPFN-GSA modifies the architecture with geospatially aware sparse attention [6], and Kriging-prior Regression feeds TabPFN kriging-derived spatial features [22]. This paper takes the opposite stance: the model as shipped, no architectural change and no spatial feature, with the dependence left for the in-context learner to find in two raw columns.

TabPFN in property valuation. To the author's knowledge, the only published applications of TabPFN to property valuation are his own, both in Portuguese. On São Paulo ITBI transaction prices, TabPFN v2 outperformed an FT-Transformer and multiple linear regression [4]; on the Belo Horizonte listings, TabPFN v3 outperformed the linear and non-linear spatial econometric models in the comparison [3]. Relative to the latter, the present study changes the target (declared prices instead of listings), the scope (three property types instead of apartments only), the scale (82,187 training rows instead of 5,005), the protocol (an out-of-time leg and the residual autocorrelation criterion are added) and the opponents (a spatially augmented XGBoost and a geographically weighted XGBoost join the econometric baselines).

3 Data

3.1 Source and cleaning

The São Paulo City Hall publishes every paid form of the real-estate transfer tax (ITBI, *Imposto sobre Transmissão de Bens Imóveis*) with the declared transaction price, the cadastral key (SQL: sector, block, lot) and the physical attributes of the property from the annual property-tax register (IPTU): built and lot area, use, construction standard and the corrected completion year (ACC, *Ano de Construção Corrigido*). The 2025 workbook holds 230,525 forms and the January–July 2026 workbook 135,655. Coordinates come from the centroids of the fiscal-block polygons of the municipal cadastre (GeoSampa), so all properties in one city block share a location; distance to the nearest subway or commuter-rail station is computed from the 203 station points of the same portal.

Table 1 gives the cleaning chain with the record count after each step; the full report, broken down by property type, is versioned with the data (`data/filter_report.md`). The cleaned bases hold 82,187 transactions for 2025 (training) and 47,810 for 2026 (test). A stratified sample of 24,000 rows of 2025 (12,000 apartments, 7,000 houses, 5,000 commercial units; "Level A") is used wherever the slower baselines must run.

3.2 Under-declaration

The ITBI tax base is max(declared price, assessed reference value), so a buyer who under-declares has no incentive to declare below the assessed value: under-declared cash deals bunch exactly at it. The diagnostics found that bunching concentrated in cash purchases, and the pipeline removes cash transactions declared within 1% of the reference value: 3,225 forms in 2025 and 1,551 in 2026 (step 11 of Table 1; the residual picture after the exclusion is in

Table 1: The cleaning chain: records remaining after each step.

#	Step	2025	2026 (Jan–Jul)
0	Raw base (all paid ITBI forms)	230,525	135,655
1	Deed nature contains "compra e venda" (sale and purchase)	207,518	124,289
2	Transmitted share = 100%	108,750	61,304
3	Cadastral status active	108,420	61,112
4	Valid 11-digit SQL key and valid date	108,420	61,112
5	Property type kept (apartment / house / commercial)	90,927	51,919
6	Built (reference) area > 0	90,927	51,919
7	Plausible completion year ($1900 \leq$ ACC $\leq$ transaction year)	90,927	51,918
8	Deduplication on (key, date, price)	90,673	51,810
9	Declared price $\geq$ R$ 10,000	89,078	51,202
10	Price / assessed reference value within $[0.3, 5]$	87,488	50,430
11	Under-declaration: cash deals pegged to the reference value excluded	84,263	48,879
12	Tukey fences (1.5 IQR) on ln(R$/m^2), per type, fences from 2025	82,193	47,814
13	Join to fiscal-block centroid (match 99.99%)	82,187	47,810

`data/underdeclaration_report.md`). Financed deals pass a lender's underwriting and are kept; Section 5.3 repeats the whole analysis on financed deals only.

3.3 Descriptive statistics

Table 2: The three bases. Unit value = declared price / built area, in R$/m^2.

Base	Rows	Apartments	Houses	Commercial	Financed	Median unit value
2025, full (training)	82,187	60,067	15,940	6,180	34%	4,836
2025, Level A (sample)	24,000	12,000	7,000	5,000	30%	4,715
2026, Jan–Jul (test)	47,810	35,354	9,429	3,027	42%	4,970

The response is strongly autocorrelated in space, as urban prices are. Moran's I of the raw log unit price on the k-NN-8 matrix used throughout the paper is 0.430 on the Level A sample ($z = 141.8$) and 0.506 on the 2026 base ($z = 235.7$): a property's unit value is close to its neighbours' far beyond chance, which is exactly the structure the spatial baselines are built to exploit. Figure 1(a) shows how the ten spatial blocks tile that structure over the city. Because coordinates are fiscal-block centroids, location is resolved to the city block rather than the address — a conservative choice shared by every model.

4 Methods

4.1 Target and features

The target is $y = \ln(\text{price/built area})$. The information set is the same for every model: built area, lot area, age, finish grade (IPTU standard level, 0–5), distance to the nearest station, month index, property type, latitude and longitude. The models differ only in what they are allowed to do with the last two.

4.2 Models

OLS hedonic. Log-linear in the areas and the distance, quadratic in age, dummies for type. The coordinates are not covariates: this is the no-location floor of the linear family.

SAR lag. $y = \rho W y + X\beta + \varepsilon$ with a row-standardised k-nearest-neighbour weights matrix ($k = 8$) built inside each training fold, estimated by the generalised-moments procedure of Kelejian and Prucha [11] as implemented in `spreg` [19]. A held-out row is predicted by the

structural form with the spatial lag taken over its k nearest *training* rows, $\hat{y}_i = \mathbf{x}_i^\top \hat{\beta} + \hat{\rho}\, \bar{y}_{N_8(i)}$, where $\bar{y}_{N_8(i)}$ is the mean observed target of those neighbours; the reduced form is not available out of sample without the test rows' own neighbourhood, so nothing from the test set enters the prediction. Ties in the centroid coordinates are broken by a deterministic sub-metre jitter (± 0.5 m, seed 42), needed because all transactions of one fiscal block share the exact centroid.

XGBoost with spatial features. The hedonic block above plus UTM coordinates, their projections on axes rotated by 30°, 45° and 60° [15], and a k-NN-8 spatial lag of the target computed inside the training fold (leave-one-out for training rows). Hyper-parameters are tuned by nested Optuna (30 TPE trials, early stopping, up to 3,000 rounds), with an inner leave-one-block-out on the training blocks; the outer test block never touches tuning [1, 5]. Three seeds. The search space and the selected values are in Appendix B.

Geographically weighted XGBoost. Geographical-XGBoost [8] fits one local XGBoost per calibration location with bi-square distance weights, $w = (1 - d^2/h^2)^2$ with h the distance to the k-th neighbour, blended with a global model as $\hat{y} = \alpha\, \hat{y}_{\text{local}} + (1 - \alpha)\, \hat{y}_{\text{global}}$. As packaged, it calibrates one local model, with its own grid search, at every training row over a dense distance matrix, which a feasibility pilot found impracticable at this scale (Appendix A). The implementation here keeps the estimator and calibrates the local models at 1,000 regression points — the K-means centres of the training coordinates, about one per 1.5 km^2 of the city — as geographically weighted regression allows [7]; a query location is predicted by inverse-distance averaging its three nearest local models. The local models see the hedonic attributes and no coordinates, so space enters only through the weighting, as in the original. The bandwidth $k \in \{50, 100, 200, 400, 800, 1600, 3200\} \times n_{\text{train}}/24{,}000$ and the blend $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ are chosen by an inner block cross-validation inside each outer fold; the grid was widened downward, from 400 to 50, after the first searches kept choosing its smallest value, and 50 neighbours is taken as the floor. Three seeds.

TabPFN-3.5, plain table. The nine raw columns, categorical type declared as such, through the hosted model (`tabpfn-client` 0.6.0, model version 3.5). Zero-shot, and a thinking-mode variant in which the fit explores configurations by internal validation on the training fold (RMSE objective, medium and high effort). No spatial information of any kind beyond the two coordinate columns. Fitting on 24,000 rows takes about 9 s; on 82,187 rows, 15 s. Settings in Appendix B; the input variants — the postal code in place of the coordinates, the free text of the form — are introduced with their results in Section 5.4.

4.3 Evaluation protocol

Leg 1: spatial block cross-validation. Ten K-means blocks on the Level A coordinates, each held out in turn [14, 20]. Every fold-internal quantity (weights matrix, spatial lag, tuning, regression points) is recomputed on the training blocks only. This asks whether a model can price a part of the city it has never seen — the harder test for a model that leans on location.

Leg 2: out-of-time. One fit on 2025 (Level A, then the full base), predictions for all 47,810 transactions of 2026, one to seven months after the training window. Nothing from 2026 touches fitting or tuning. Same city, so the spatial-lag models get neighbours from the same streets: the most favourable setting for explicit spatial modelling. Leakage check: no transaction (key, date, price) appears in both years; 0.6% of 2026 rows share a cadastral key with Level A.

Metrics and tests. RMSE, MAE and R^2 in log units, MAPE in price units, mean residual (bias). Residual spatial autocorrelation by Moran's I [16] on a k-NN-8 matrix of the test coor-

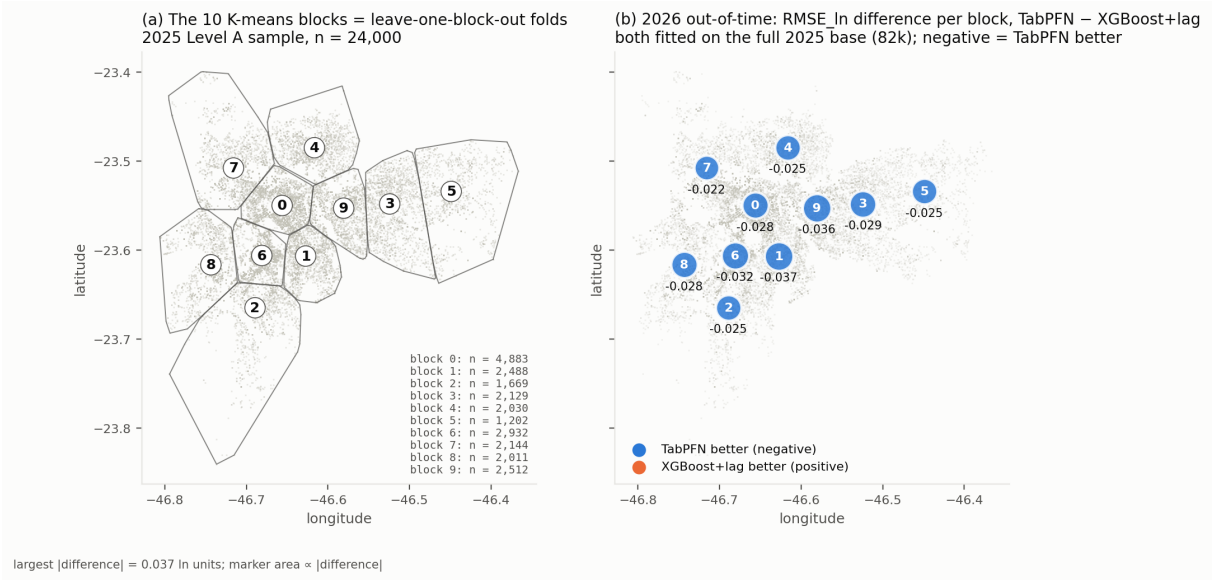


Figure 1: (a) The ten K-means blocks of the Level A sample, each held out in turn. (b) Per-block RMSE difference TabPFN-3.5 − XGBoost+lag on the 2026 test, both fitted on the full 2025 base; negative in every block.

dinates. Paired comparisons over the ten blocks (for 2026, each transaction is assigned to the block of its nearest 2025 K-means centroid): exact Wilcoxon signed-rank test on the per-block RMSE and a block bootstrap (2,000 resamples of blocks) for the pooled difference with a 95% percentile interval. Where a model is stochastic, three seeds (42, 43, 44) and the per-seed range; a TabPFN run (seed 42) is paired against every seed of a stochastic opponent.

5 Results

5.1 Leg 1: unseen blocks

Table 3: Leave-one-block-out CV on Level A (24,000 rows of 2025). Out-of-fold metrics in log unit price; Moran's I of the out-of-fold residuals on a k-NN-8 matrix. Point values are seed 42; parenthesised ranges are over three seeds.

Model	RMSE	MAPE	R^2	Moran's I
Median by type (floor)	0.497	42.1%	−0.04	0.446
OLS hedonic	0.470	39.4%	0.066	0.390
SAR lag, GM ($\hat{\rho} = 0.91$)	0.448	38.4%	0.155	0.363
XGBoost, rotated coordinates, no lag	0.397	33.3%	0.334	0.355
XGBoost, rotated coordinates + k-NN-8 lag	**0.385** (0.385–0.387)	32.3%	**0.376**	**0.309** (0.309–0.318)
Geographically weighted XGBoost	0.410 (0.407–0.410)	34.5%	0.291	0.324 (0.319–0.326)
TabPFN-3.5, plain table, zero-shot	0.394 (0.391–0.394)	32.2%	0.346	0.348 (0.342–0.348)
TabPFN-3.5, plain table, thinking (medium)	0.389	**31.7%**	0.362	0.342
TabPFN-3.5, plain table, thinking (high)	0.388	**31.7%**	0.366	0.339

TabPFN-3.5 zero-shot beats the SAR lag model in 8 of 10 blocks (ΔRMSE = −0.054, 95% CI $[-0.100, -0.014]$, Wilcoxon $p = 0.010$; ΔMAPE = −6.2 pp). Against XGBoost with the explicit lag the difference is a tie: ΔRMSE = +0.009 $[-0.018, +0.034]$, 7 blocks won, one clear loss in the largest block. Against the geographically weighted XGBoost it is also a tie, leaning TabPFN's way: ΔRMSE −0.013 to −0.016 across that model's three seeds, every interval across zero, 6 or 7 blocks won. The two XGBoost variants are not interchangeable: the geographically weighted one loses to XGBoost+lag by +0.023 to +0.025 (1 or 2 blocks won, $p = 0.027$–0.037), so the fully spatial global XGBoost is the hardest opponent in this leg. Thinking mode lowers

the zero-shot error by a small margin (-0.005 $[-0.008, -0.001]$ against seed 42, 7 of 10 blocks, though the exact Wilcoxon does not reach significance, $p = 0.105$; about -0.002 against the other seeds) and gives the lowest MAPE in the table; high effort costs about 13 times more per fold and adds nothing beyond noise.

On the residual criterion the hypothesis is partly refuted in this leg. Moran's I of the plain-table model (0.348) lies below SAR (0.363) but above both XGBoost variants: the global one with the explicit lag (0.309) and the geographically weighted one (0.319–0.326). Inside XGBoost, the lag alone is worth $\Delta\text{RMSE} = -0.013$ $[-0.024, -0.004]$ and lowers the residual I from 0.355 to 0.309 — the measurable value of modelling space explicitly when the test is a part of the city no model has seen. The refutation is partial and on the record.

Errors are large in absolute terms (R^2 of 0.29–0.38 for the non-linear models, against 0.538 in the listings study): declared prices, three property types and centroid coordinates all add noise, and every model leaves spatial structure behind in this leg.

5.2 Leg 2: one year ahead

Table 4: Out-of-time test: fit on 2025, predict the 47,810 transactions of 2026. Bias is the mean residual in log units (positive = under-prediction). Moran's I on a k-NN-8 matrix of the 2026 coordinates. Point values are seed 42; ranges are over three seeds.

Training base	Model	RMSE	MAPE	R^2	Bias	Moran's I
Level A (24k)	OLS hedonic	0.404	32.4%	0.185	+0.033	0.408
	SAR lag	0.346	27.0%	0.401	+0.032	0.224
	XGBoost + lag	0.288	21.8%	0.584	+0.047	0.138–0.140
	Geographically weighted XGBoost	0.313 (0.313–0.314)	24.0%	0.508	+0.045–+0.048	0.187–0.188
	TabPFN-3.5, zero-shot	**0.275**	**20.4%**	**0.622**	+0.055	**0.122**
	TabPFN-3.5, thinking (medium)	0.275	20.5%	0.621	+0.055	0.121
Full (82k)	OLS hedonic	0.401	32.4%	0.194	+0.023	0.397
	SAR lag	0.326	25.0%	0.470	+0.031	0.134
	XGBoost + lag	0.294[a]	21.5%[a]	0.568	+0.103[a]	0.107[a]
	Geographically weighted XGBoost	0.298[b]	22.1%	0.557	+0.065[b]	0.163[b]
	TabPFN-3.5, zero-shot	**0.265**	**19.5%**	**0.649**	+0.058	**0.092**

[a] seed 42; three seeds: RMSE 0.283–0.294, MAPE 20.9–21.5%, bias +0.070–+0.103, I 0.103–0.107. [b] seed 42; three seeds: RMSE 0.298–0.301, bias +0.065–+0.078, I 0.163–0.166.

Paired over the ten 2026 blocks, TabPFN-3.5 zero-shot trained on Level A beats XGBoost with the explicit lag in 10 of 10 ($\Delta\text{RMSE} = -0.013$ $[-0.016, -0.011]$, $\Delta\text{MAPE} = -1.4$ pp, $p = 0.002$), the geographically weighted XGBoost in 10 of 10 (-0.038 to -0.039 across its seeds, every interval within $[-0.049, -0.028]$) and SAR by -0.071 $[-0.079, -0.062]$. On the full base the gap to XGBoost+lag widens to -0.029 $[-0.032, -0.026]$ ($\Delta\text{MAPE} = -2.0$ pp, again 10 of 10, and by -0.018 to -0.020 against its two better seeds); against the geographically weighted XGBoost it is -0.033 to -0.036 with 9 of 10 blocks ($p = 0.004$; the one loss is the smallest block, 2,425 transactions, by 0.001–0.002), and the plain-table model has the lowest residual autocorrelation of the five models. Scaling from 24k to 82k rows improves TabPFN by -0.010 $[-0.012, -0.008]$ and SAR by -0.020; XGBoost+lag does not improve ($+0.006$ $[+0.001, +0.011]$ at seed 42) because its trend bias more than doubles, from $+0.047$ to $+0.103$, and eats the variance gain. The bandwidth of the geographically weighted model hardly matters here: with 400, 200, 100 or 50 neighbours its 2026 RMSE on Level A stays between 0.308 and 0.314. Errors are flat across blocks (0.23–0.31) and across the monthly horizons (0.26–0.28 over the months with at least 500 transactions): no subset carries the result. Thinking mode adds nothing in this leg. The hypothesis survives every criterion, at both scales, against all three spatial specialists.

The limitation worth stating. Every model under-predicts 2026, a rising market, and the non-linear ones more (bias +0.04 to +0.10 in ln against +0.02 to +0.03 for OLS and SAR): trees and TabPFN hold the last observed level of the month index instead of extrapolating a trend (Figure 4). Appraisal practice would apply a price index; it is reported here, not corrected.

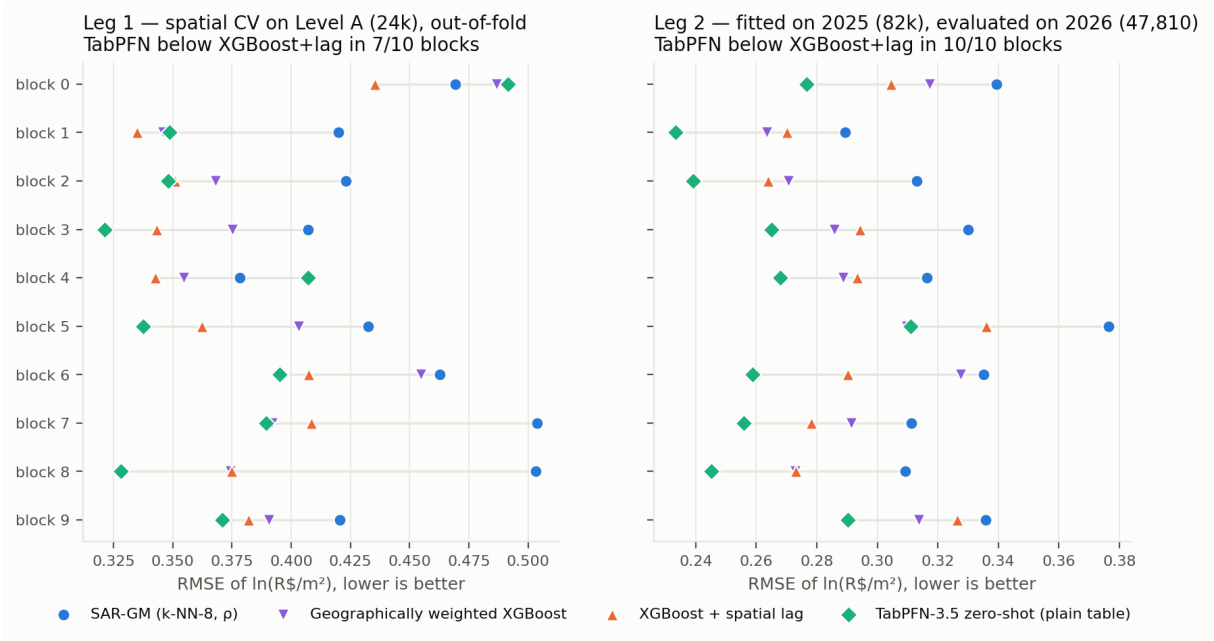


Figure 2: RMSE per block. Left panel: leave-one-block-out CV on Level A (leg 1), where TabPFN-3.5 ties XGBoost+lag; right panel: the 2026 out-of-time test with models fitted on the full 2025 base (leg 2), where it is below XGBoost+lag in every block.

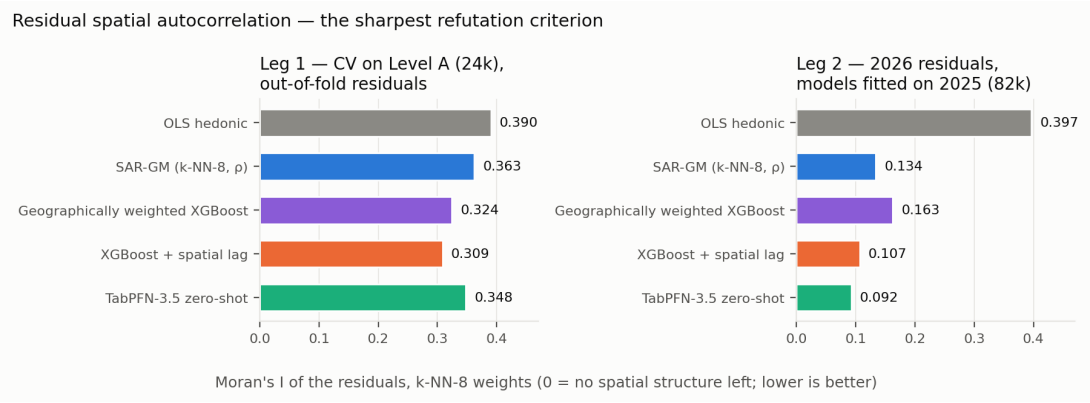
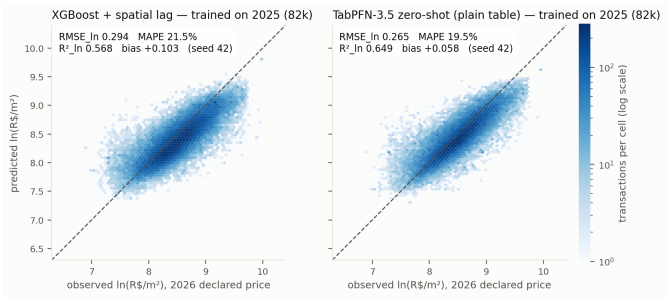


Figure 3: Residual Moran's I, the sharpest refutation criterion. Left panel: the block CV (leg 1), where the plain-table model leaves more spatial structure than either XGBoost — the partial refutation; right panel: 2026 (leg 2), where it leaves the least of the five models.

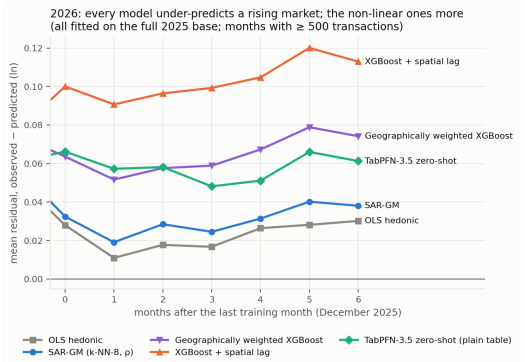
5.3 Robustness: financed transactions only

Financed transactions carry a bank appraisal and are the cleanest prices in the base. Both legs were re-run on them alone: 7,211 of the 24,000 Level A rows for the block CV, the same rows fitted once and scored on the 20,052 financed transactions of 2026.

Cleaner prices lower every error and leave the one-year-ahead ranking unchanged. In the block CV the plain-table model moves ahead of XGBoost+lag (ΔRMSE $= -0.017$ $[-0.032, -0.003]$, $p = 0.049$, 8 of 10 blocks against its seed 42; 9 of 10 against seeds 43 and 44) and the partial refutation on the residual criterion does not reappear: it leaves less spatial structure than the spatial XGBoost in both legs (0.279 against 0.327–0.340 in the CV). The headline results do not depend on the cash deals retained by the under-declaration filter; if anything, the noisier cash prices are where the explicitly spatial learner holds its own. The geographically weighted XGBoost was not re-run on this subset.



(a) Predicted vs observed, 2026.



(b) Mean residual by month ahead.

Figure 4: The rising market of 2026: every model under-predicts, the non-linear learners more so.

Table 5: Both legs on financed deals only (seed 42).

Model	CV RMSE	CV MAPE	CV I	2026 RMSE	2026 MAPE	2026 I
OLS hedonic	0.354	28.5%	0.343	0.343	26.5%	0.491
SAR lag	0.344	28.1%	0.342	0.286	22.0%	0.297
XGBoost + lag	0.286	23.0%	0.327	0.221	16.7%	0.160
TabPFN-3.5, zero-shot	**0.269**	**21.1%**	**0.279**	**0.207**	**15.3%**	**0.143**

5.4 What carries location

The main analysis hands the model two numbers. The ITBI form also states location as names — the postal code and a free-text district field — and describes the property in words. Four input variants, run under the identical protocol with the baselines untouched, ask what the model is actually reading. Each was pre-registered with an expectation in the module docstring of `src/model_tabpfn.py` before the run; the expectations, and where they failed, are reported with the results.

The postal code in place of the coordinates. The CEP (*Código de Endereçamento Postal*) is Brazil's ZIP code: eight digits, in São Paulo usually one street or a stretch of one, 10,022 distinct values in the 24,000 rows of Level A and 15,938 in the full 2025 base, and nothing in the column says which code lies next to which. The variant removes latitude, longitude and the station distance (computed from them), and puts the raw code in their place as a string — the only spatial information the model receives, against baselines that keep every spatial input. The written expectation was that the code would recover less than half of the way from the no-location floor to the coordinate model and lose to XGBoost+lag in both legs.

Table 6: The postal code in place of the coordinates (zero-shot, seed 42) against the plain table. RMSE in log unit price · Moran's I of the residuals; plain-table leg-1 entries are ranges over three seeds. The baselines are those of Tables 3 and 4.

TabPFN-3.5 input	Leg 2, 24k → 2026	Leg 2, 82k → 2026	Leg 1 (block CV)
Postal code only	0.277 · 0.129	0.268 · 0.091	0.457 · 0.502
Latitude and longitude	0.275 · 0.122	0.265 · 0.092	0.391–0.394 · 0.342–0.348

One year ahead, the expectation failed in the direction favourable to the model: the label does the work of the coordinates. The postal-code-only model recovers 94–97% of the way from its no-location floor (0.352 on Level A, 0.327 on the full base) to the coordinate model and beats every seed of every spatial baseline: XGBoost+lag by −0.011 (24k) and −0.014 (82k) against its best seed, in 10 of 10 blocks; the geographically weighted XGBoost by −0.036 and −0.029 (10 and 9 of 10); SAR by −0.069 and −0.057 (10 of 10; Wilcoxon $p = 0.002$ throughout, $p = 0.004$ for the 9-of-10 row). It also leaves less residual autocorrelation than all three — and the Moran's

I is computed on a k-NN-8 matrix of the very coordinates the model was denied. The result is not a price lookup by code: on the 2026 transactions whose code never occurs in the training base (22% against Level A, 9% against the full base), the model still recovers 97% and 96% of the location signal, about as much as on the codes it has seen (98% and 94%). Nor is the code richer information than the baselines had: inside TabPFN it comes within 0.004 of the coordinate pair and, added on top of the coordinates, it changes nothing (below); XGBoost had the coordinates *and* a lag, rotated axes and the station distance.

Where the label stops is a part of the city never seen. In leg 1 a held-out block's codes are absent from training by construction (97–100%), and there the code carries worse than nothing: the model lands below its own no-location floor (0.457 against 0.440), loses to XGBoost+lag on pooled error (+0.069 to +0.072, 4 of 10 blocks), and leaves the most spatial autocorrelation of any model in the study ($I = 0.502$) — a whole block of unknown codes is priced as one unknown, and the error shifts as a block. This leg behaved as the written expectation said, and worse. Coordinates extrapolate into an unseen district; a label cannot. Why codes never seen among known ones still work is not settled; the leading hypothesis is that Brazilian postal codes are assigned geographically, so neighbouring codes mostly mean neighbouring streets, and the test that would decide it (a fixed random bijection of the codes, destroying the order while keeping the identities) is designed but has not been run.

The address on top of the coordinates. Adding the postal code and the district field to the plain table, as filed, moves the RMSE by less than 0.006 in either direction in every test and combination, the MAPE by at most 0.3 points, and the residual Moran's I by at most 0.019 in the block CV and 0.008 one year ahead; the margin over XGBoost+lag stays where it was. Two raw numeric columns already carry what the address carries. One side observation is reported without an explanation: with both names in, the mean under-prediction of 2026 drops from +0.055 to +0.045 in ln on Level A and from +0.058 to +0.043 on the full base, while the RMSE stays put.

The unit and the building as written. The form's unit complement (`AP 201 E 3VGS`: apartment 201 with three parking spaces) and its *Referência* field, which mostly names the building (`EDIFICIO THE PARK`), go in as raw strings next to the coordinates; nothing is parsed. The written expectation was a small gain at most, one year ahead, where the same buildings sell again, with leg 1 as a near-negative control. The expectation was wrong about where.

Table 7: The unit and building text added to the plain table (zero-shot, seed 42). Δ is text − plain, block-bootstrap 95% CI; blocks won by the text variant.

	Plain table	+ unit and building text	ΔRMSE [95% CI] · blocks
Leg 1, block CV	0.3938 · 32.2% · I 0.348	0.3835 · 30.8% · 0.331	−0.0103 [−0.0161, −0.0031] · 8/10
Leg 2, 24k → 2026	0.2750 · 20.4% · 0.122	0.2750 · 20.6% · 0.122	+0.0000 [−0.0011, +0.0010] · 6/10
Leg 2, 82k → 2026	0.2649 · 19.5% · 0.092	0.2610 · 19.2% · 0.086	−0.0039 [−0.0049, −0.0031] · 10/10

The largest gain is in the district never seen, against each of the three seeds of the plain table (−0.007 to −0.010, every interval clear of zero). There the text turns the tie with the geographically weighted XGBoost into a win (−0.023 to −0.026, 9 of 10 blocks, $p = 0.027$), keeps a tie leaning TabPFN's way against XGBoost+lag, and gives the lowest MAPE of any model in that leg (30.8%). It is not string matching: 92% of a held-out block's building names never occur in the training folds, and complements never seen as exact strings gain as much as those seen (−0.014 against −0.013). The gain sits in forms that mention parking (−0.034), commercial units (−0.023) or a named building (−0.017): what transfers to an unseen district is what the words say, not which building they point to. The residual criterion narrows but stands: Moran's I falls from 0.348 to 0.331, still above both XGBoost variants.



(a) Mean |SHAP| by feature and type.

(b) One valuation, base value to estimate.

Figure 5: Permutation SHAP on 50 transactions of 2026 (independent masker, 20-row background), model reloaded from its cached record. Values in log units; e^{SHAP} is the multiplier on the unit value.

Thinking with the blocks as groups. Thinking mode can split its internal validation by a grouping column; with the spatial block as the group, the internal validation holds out whole districts, as leg 1 does. Against thinking without groups the change in leg 1 is $+0.0038$ $[-0.0055, +0.0149]$ (3 of 10 blocks) and the residual I moves from 0.342 to 0.350: grouping does not close the gap. One year ahead the change is -0.0014 $[-0.0021, -0.0007]$ (9 of 10 blocks) and it cuts the mean under-prediction from $+0.055$ to $+0.043$, but there the block label also tells the model which part of the city a 2026 row belongs to, so the small gain is not purely a tuning effect. On an unseen district the limit is information, not validation design: the text moves leg 1, the grouping does not.

5.5 Explaining an estimate

An appraiser who signs a valuation has to say why. The model is served through an API, so the explanation is model-agnostic permutation SHAP [13] on 50 stratified 2026 transactions the model had never seen, with every one of the 27,217 predicted rows cached for offline reproduction. Reloaded from its cached record, the model reproduces the predictions it is explaining (RMSE 0.296 in ln on the 50 rows), so the attributions describe the run that was scored. Longitude (mean |SHAP| 0.184, a typical $\times 1.20$ on the unit value) and latitude (0.143, $\times 1.15$) are the two strongest inputs, ahead of property type (0.127), built area (0.112) and age (0.082). Directions match appraisal experience: age and distance to a station pull the unit value down, finish grade pushes it up, larger built areas lower the price per square metre. The two raw coordinates carry, inside the model, the signal the baselines encode with a weights matrix — and Section 5.4 shows that what they carry, within an observed market, is location as an identifier rather than geometry.

6 Discussion

Read together, the two legs draw one line. Inside a market the model has seen — the same city, one to seven months ahead — in-context learning over 24,000 to 82,000 rows recovers the neighbourhood structure from two raw columns and prices better than the models built to encode it, by every pre-registered criterion, residual autocorrelation included. The postal-code variant sharpens the reading: within that market the model does not need geometry at all, only a fine-grained location identifier, and it prices codes it has never seen among known ones almost as well as known codes. Across the line, in a district with no sales in the training base, the identifier collapses, the coordinates still extrapolate, and the explicit spatial lag earns its keep on the residual criterion: the partial refutation of leg 1 is the measured price of not modelling space when the test block is new ground. That price shrinks with more information — the free text

of the form cuts the gap and the error — and does not appear on the lender-audited financed prices against XGBoost+lag; it does not vanish in the main analysis, and it is reported as found.

The one systematic error shared by all models is the trend. Every model under-predicts the rising market of 2026, the non-linear learners more (+0.04 to +0.10 in ln against +0.02 to +0.03 for the linear models), because trees and the foundation model hold the last observed level of the month index instead of extrapolating it. Appraisal practice corrects this with a price index as a matter of routine; no correction was applied here, so that the comparison stays between models as fitted. The bias concerns the level rather than location: the residual maps and Moran's I, which the trend shifts uniformly, still separate the models.

The engineering ledger is short. The SAR model needs a weights matrix per fold; the spatial XGBoost needs a fold-internal lag, rotated axes and 30 nested tuning trials (about 15 minutes of tuning on the full base); the geographically weighted XGBoost needs 1,000 local models and an inner search over bandwidth and blend (about ten minutes). TabPFN-3.5 needs the table: one fit of 9 to 15 seconds, and, where a postal code is already on the form, not even a geocoder. For mass appraisal under the Brazilian standard NBR 14653-2, two consequences follow. The standard's precision grades are defined on interval estimates, and the model returns predictive quantiles at no extra engineering — they are cached for every zero-shot fit in this study, and their coverage analysis is the natural next step. And the standard's demand that a valuation be grounded can be met in part by model-agnostic attribution: the SHAP directions above are the ones an appraiser would defend.

The limits of the evidence are these. Coordinates are fiscal-block centroids, so location is resolved to the city block and all transactions of one block coincide; a sub-metre deterministic jitter breaks ties for the weights matrices only. Prices are self-declared; the under-declaration filter removes the detectable bunching, not the phenomenon, and the financed-only robustness bounds its effect. The study covers one city and one period, and the foundation model is served through an API: its internals are not inspectable (hence no exact per-row attribution for the regression head), and the hosted version can change, though the weights are published for research use and every API response is cached. The TabPFN input variants ran on a single seed; the plain-table seed spread in leg 1 (0.003 in ln) is far below every leg-2 margin, but the leg-1 ties should be read with that width in mind. The geographically weighted XGBoost is an approximation of the published estimator — local models at 1,000 regression points rather than at every training row, shared local hyper-parameters, a bandwidth floor of 50 neighbours that the inner search chose in 8 of 10 folds — defensible on geographically weighted regression's own terms, and bounded by the bandwidth sensitivity reported above. The shuffled-code test that would settle why unseen postal codes work is designed, not run.

7 Conclusion

The hypothesis was that a tabular foundation model receiving coordinates as two plain numeric columns predicts spatially correlated prices as well as, or better than, models that encode spatial dependence explicitly, and it was put at risk against three such specialists — SAR, XGBoost with a spatial lag and a geographically weighted XGBoost — on 130,000 declared transaction prices, with the refutation criteria written first. One year ahead, inside the observed market, the hypothesis survives every criterion at both training scales: lower pooled error than SAR, XGBoost with a spatial lag and a geographically weighted XGBoost, wins in at least 9 of 10 spatial blocks against every seed of every opponent, and the least residual spatial autocorrelation of the five models — a verdict that stands even when the coordinates are replaced by a raw postal code. In a district never seen, it ties the two XGBoost variants on error, beats SAR, and is partly refuted on the residual criterion, leaving more spatial structure than either XGBoost; the free text of the tax form narrows that gap and the financed-only subset dissolves it against XGBoost+lag, but in the main analysis it stands. The verdict could change elsewhere: another city or market,

listings instead of declared prices, a model version with row-level attribution for regression, or the shuffled-code test, are the experiments that would move it. As it is, within a market that has a sales history, none of the spatial machinery tested here improved on the raw table; its remaining value, measured, is extrapolation to ground without sales.

Reproducibility and data availability

Code, cleaned data, cached model responses and every number in this paper are at `https://github.com/abenvenho/tabpfn-itbi-sp` (Apache-2.0); all source data is public (municipal open data under the Brazilian access-to-information law). The pipeline was re-run from a clean clone: the three bases were rebuilt byte-identical from the raw public workbooks, and the smoke and baseline runs returned the documented numbers. Every number cited in this manuscript is checked against the versioned results by `paper/check_numbers.py`.

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A Geographically weighted XGBoost: feasibility pilot

The published implementation (`geoxgboost`) calibrates one local XGBoost, with its own grid search, at every training row over a dense $n \times n$ distance matrix. A pilot on 200 rows with one bandwidth candidate took 18.7 s (0.093 s per spatial unit), which extrapolates to about half an hour per bandwidth candidate per fold at the 19,117 training rows of a Level A fold, about two days for ten candidates over ten folds, and 2.9 GB of distance matrix per fold on the study machine (2 vCPU, 7 GB RAM; `results/gxgb_timing_pilot.json`). The regression-point approximation of Section 4 keeps the estimator — local models, bi-square weights, blend with a global model, nested bandwidth choice — at about one minute per fit and ten per tuning.

B Hyper-parameters and model settings

XGBoost + spatial features. Nested Optuna, 30 TPE trials per outer fold, inner leave-one-block-out on the training blocks, early stopping with up to 3,000 rounds. Search space: learning rate 0.01–0.2 (log), max depth 3–10, min child weight 1–50 (log), subsample 0.5–1, colsample_bytree 0.4–1, λ 10^{-3}–30 (log), α 10^{-3}–10 (log), γ 10^{-4}–1 (log). Selected for the full-base out-of-time fit (seed 42): learning rate 0.049, depth 7, min child weight 2.0, subsample 0.89, colsample 0.56, λ 0.0033, α 0.0041, γ 0.0031, 398 rounds; inner RMSE 0.351; tuning time 901 s (`results/xgb_params_*.json`).

Geographically weighted XGBoost. $M = 1{,}000$ regression points (K-means of the training coordinates); local models with shared hyper-parameters (300 rounds, learning rate 0.05, depth 4); global model with 800 rounds, learning rate 0.05, depth 6; a query predicted by the inverse-distance average of its 3 nearest local models, blended by α. Bandwidth k and α by nested inner block CV, grids in Section 4 (the grid values scale with the training size, hence 40–176 on the 19–21k-row training folds of Level A). Chosen: on Level A, k per fold between 40 and 176 (leg 1) and $k = 50$ (leg 2); on the full base, $k = 342$; $\alpha = 0.5$ throughout (`results/gxgb_params_*.json`).

TabPFN-3.5. Hosted model, `tabpfn-client` 0.6.0, `create_default_for_version("v3.5")`, regression head, zero-shot unless stated. Thinking mode: `thinking_effort` medium or high, `thinking_metric = "rmse"`, timeout 2,400 s; the grouped variant passes the spatial block as `group_col`. The 82,187-row fits set `ignore_pretraining_limits`. Predictive quantiles are requested and cached on every zero-shot fit. Every API response is cached under `results/tabpfn_cache/`, so the full protocol re-runs offline from the cached predictions.